

Tactica

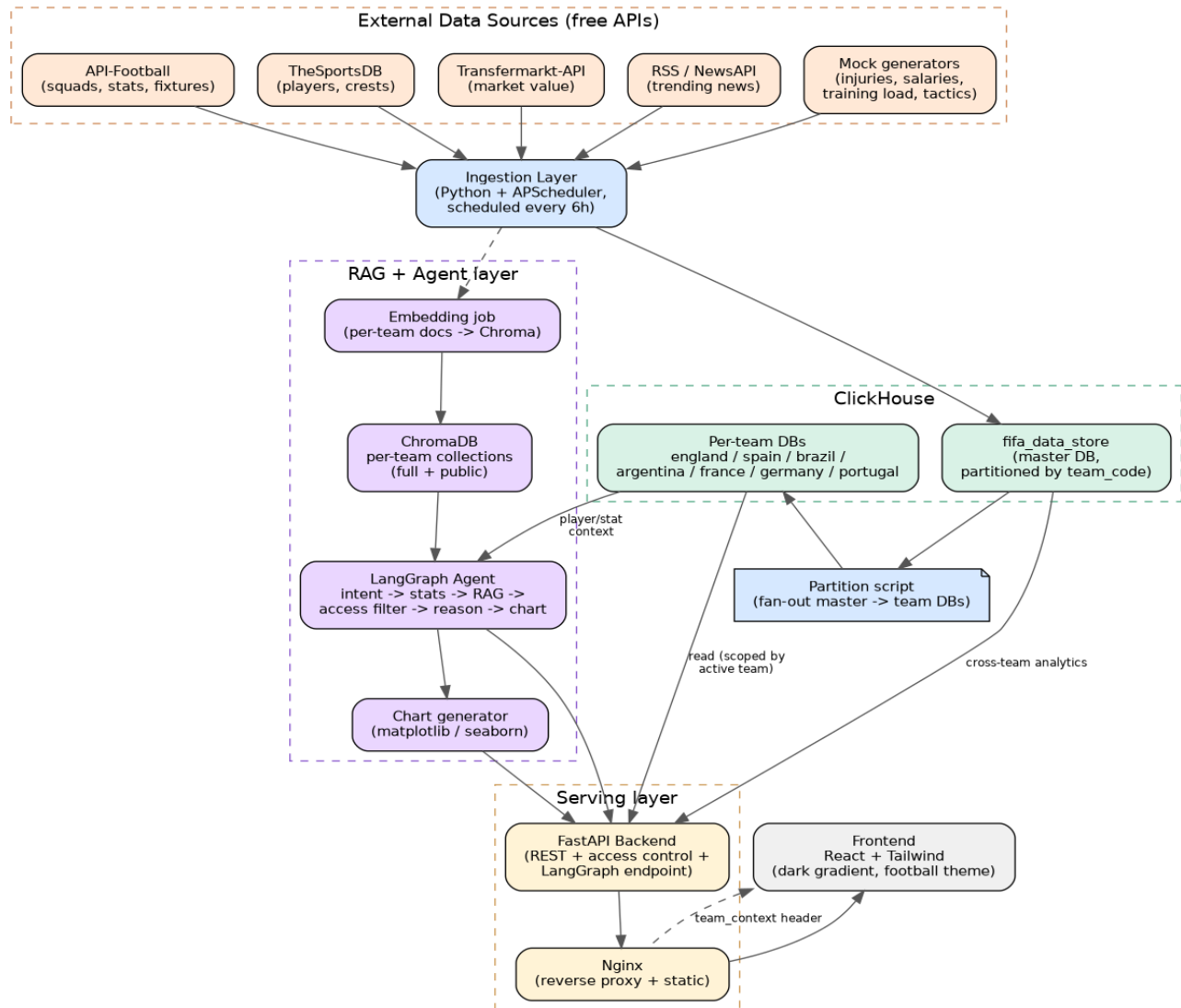
FIFA World Cup Manager Console — Architecture & Design Reference

This document contains the high-level architecture, low-level architecture (data flow and container deployment), UI wireframes, and design system specification for the Tactica project: a RAG + LangGraph powered FIFA World Cup manager dashboard and tactical assistant, built on ClickHouse, ChromaDB, FastAPI, React/Tailwind, and Docker.

Companion document: [TACTICA_BUILD_SPEC.md](#) (step-by-step build instructions for AI coding agents)

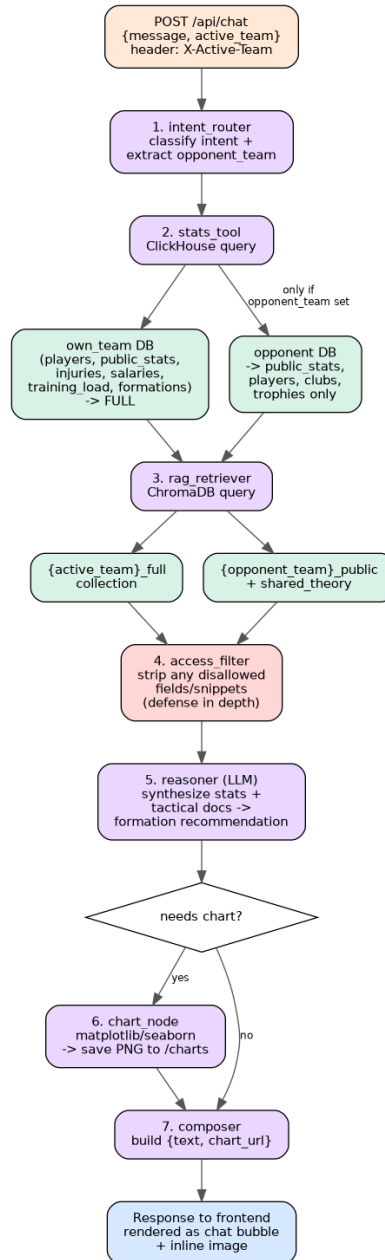
1. High-Level Architecture

Data flows from free external APIs and synthetic mock generators into an ingestion layer, which loads a master ClickHouse database (fifa_data_store) partitioned by team. A partitioning script fans this out into seven isolated per-team databases. The RAG + agent layer embeds per-team knowledge base documents into ChromaDB and uses a LangGraph agent to answer tactical questions, generating charts on demand. FastAPI exposes everything through an access-controlled REST API, served via Nginx to a React/Tailwind frontend.



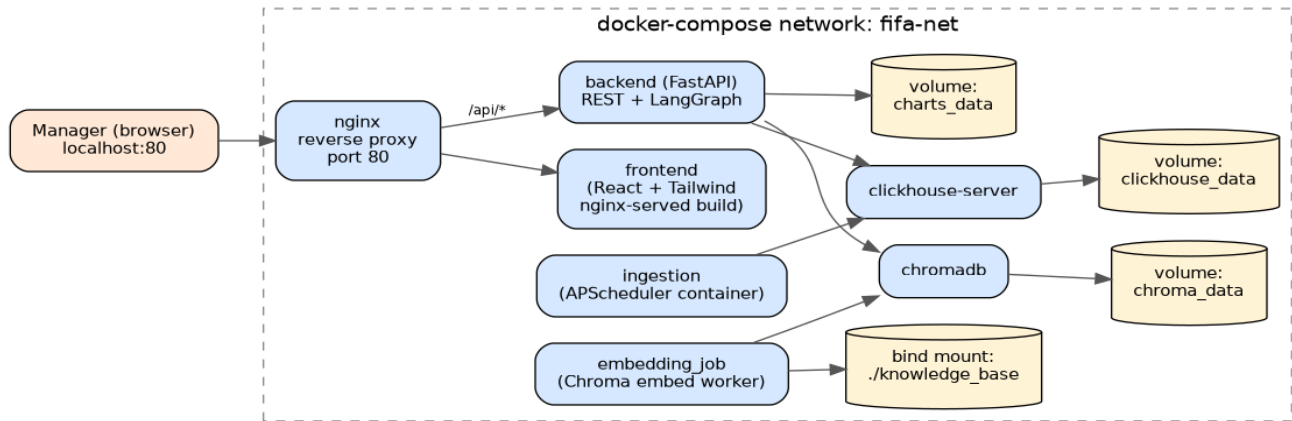
2. Low-Level Architecture — Chatbot Request Flow (LangGraph)

Every chatbot message runs through a 7-node LangGraph pipeline. The `access_filter` node (step 4) is the enforcement point that guarantees opponent-team private data (injuries, salaries, training load) is never passed to the reasoning LLM — it re-applies the same access control matrix used by the REST 'inspect' endpoints, so the privacy rule has a single source of truth.



3. Low-Level Architecture — Container & Persistence Layout

All services run under a single docker-compose network. ClickHouse and ChromaDB data live in named Docker volumes; generated chart PNGs live in a third named volume; the knowledge_base folder is bind-mounted so tactical notes can be edited live and re-embedded. Running 'docker compose stop' and 'start' preserves all volumes — only 'docker compose down -v' removes them.

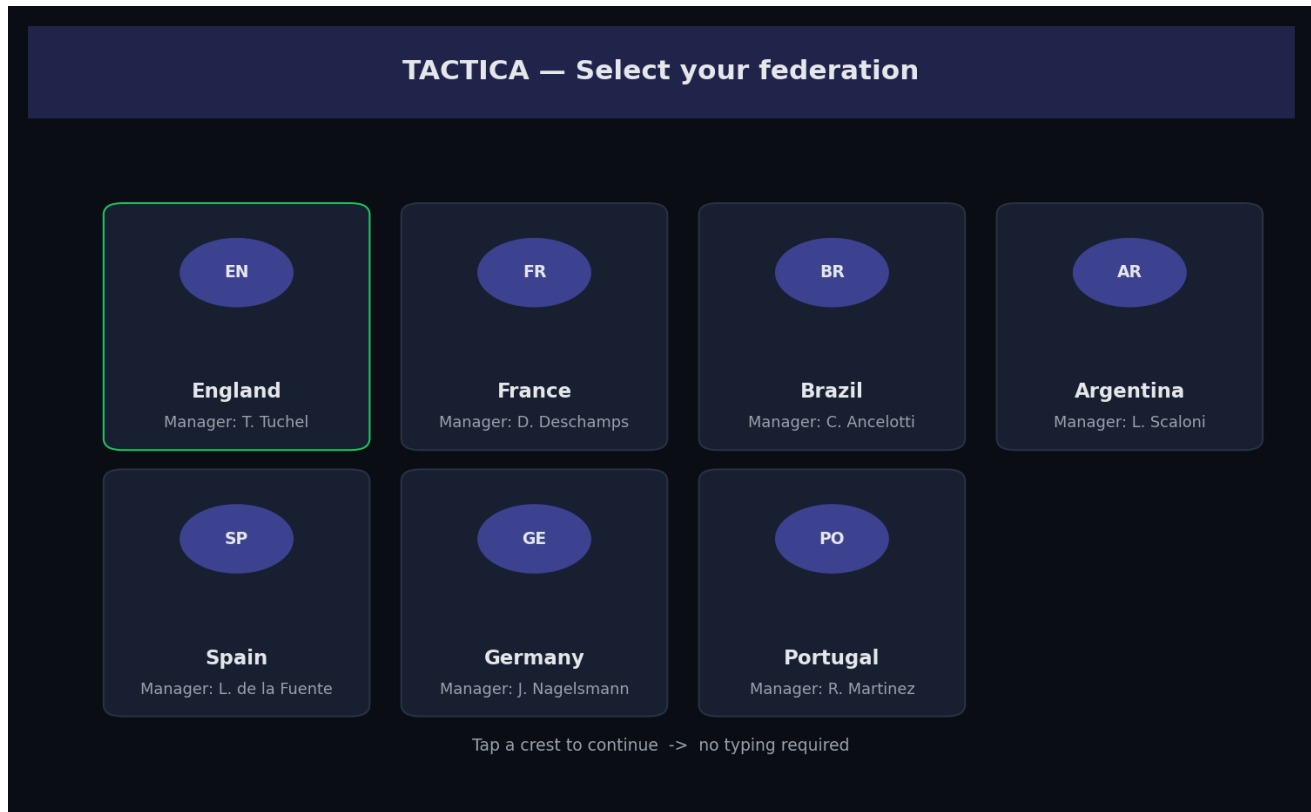


Access Control Matrix

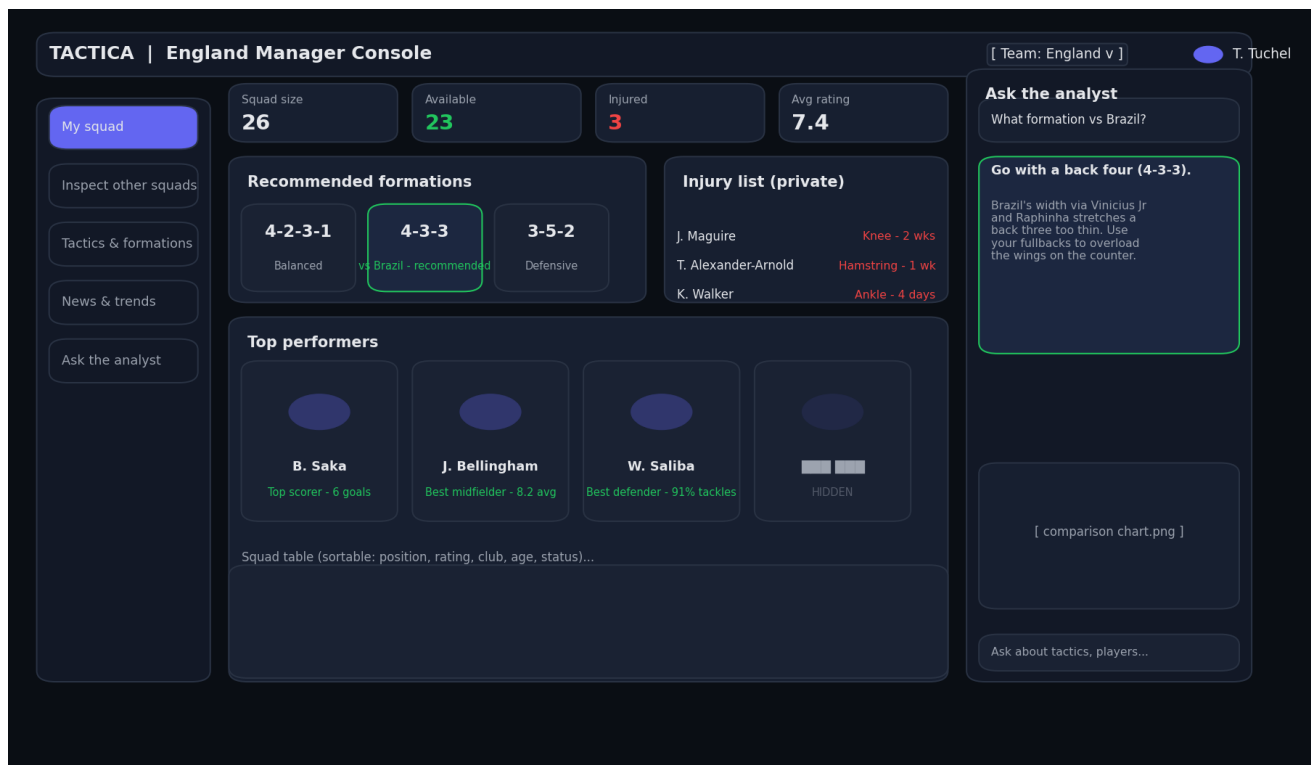
Table	Active team (own data)	Other team (inspect mode)
players (basic fields)	Full	Full
public_stats	Full	Full
clubs / trophies / matches	Full	Full
injuries	Full	Blocked -> null
salaries	Full	Blocked -> null
training_load	Full	Blocked -> null
formations	Full	name + suitable_vs only

4. UI Wireframes

4.1 Login page — select federation (no text input)



4.2 Dashboard — own squad (full access)



4.3 Inspect other squads — redacted private fields

Same layout as the dashboard, rendered for an opponent team. Injuries, salaries and training load render as blurred placeholder bars with a lock icon and the caption 'Private to {team} staff' — the backend sends these fields as null, the frontend never receives real values.

TACTICA | Inspecting: Brazil (logged in as England)

[Inspect: Brazil v] T. Tuchel

Read-only public view -- injuries, salaries and training load are hidden for other federations

Top performers (public)

Vinicius Jr

Top scorer - 9 goals

Raphinha

Best winger - 7.8 avg

Marquinhos

Best defender - 88% tackles

Squad (basic info + public stats)

Player	Position	Club	Rating
Vinicius Jr	LW	Real Madrid	8.9
Raphinha	RW	Barcelona	8.4
Casemiro	CDM	Man Utd	7.6
Alisson	GK	Liverpool	7.9

Trophies & history (public)

- World Cup 2002

- Copa America 2021

- Confederations Cup 2013

Injury list

lock

Private to Brazil staff

Salaries & contracts

lock

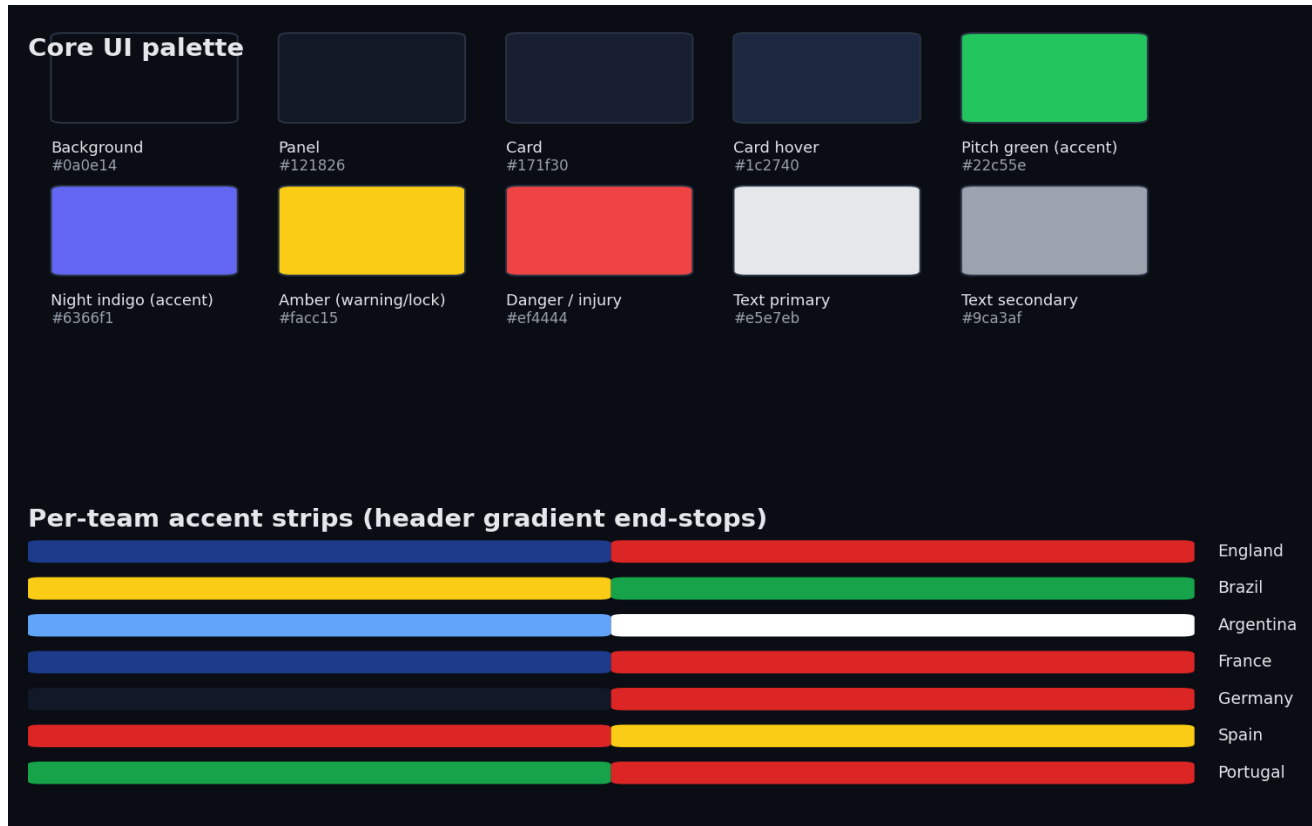
Private to Brazil staff

Training load

Private to Brazil staff

5. Design Expectations

Dark, gradient, football-themed UI. Flat dark surfaces (near-black background, slate panels and cards), with two recurring accents: a pitch-green for positive/recommended states and a night-indigo for identity/avatars/active nav. Each team has its own accent pair drawn from its kit colors, blended subtly into the header gradient when that team is active. Locked/private content always uses the amber lock motif, never just disappears.



Key UI rules

- No text input on the login page — selection is via crest tap only.
- Team switch (header dropdown) triggers a full page reload to refresh team_context everywhere.
- Private fields are returned as null by the API and rendered as blurred bars + lock icon, never omitted from the payload.
- Recommended formation card gets a 2px accent border + colored caption, others stay neutral.
- Chat panel is persistent on the right of the dashboard; chart images render inline as PNGs.
- Cards: 12px radius, 1px borders (#2a3344), no heavy shadows, generous padding.